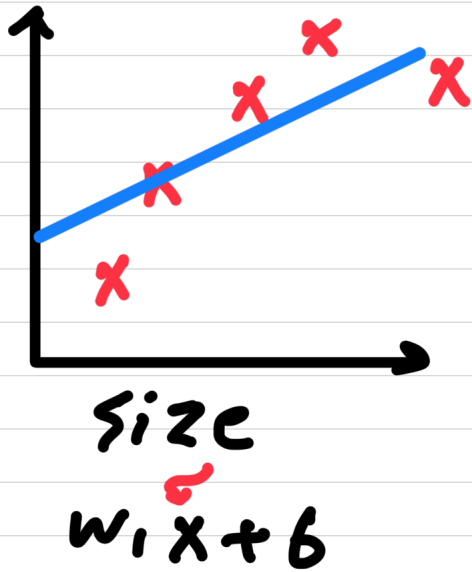


Regression ex:

Case 1



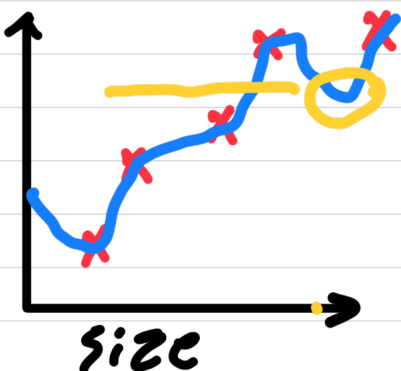
- the model **doesn't fit** the training data well
underfit (High bias)

Case 2



- the model **fits** the training data well
generalization
Best one ✓

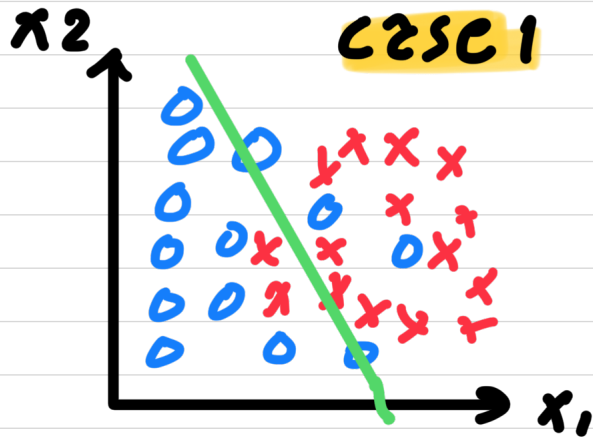
Case 3



$$w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4 + b \dots$$

- fits the training data **extremely** well
overfit (high variance)

Classification ex

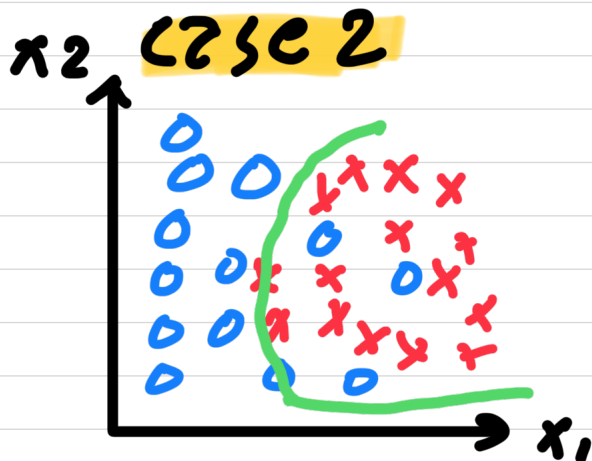


• It looks like good fit to the data

undersfit (high bias)

$$z = w_1 x_1 + w_2 x_2 + b$$

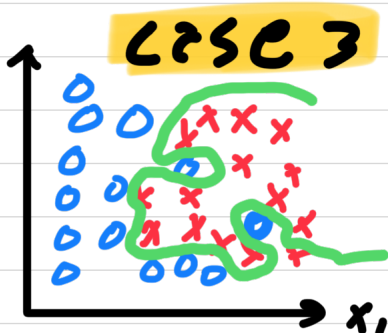
$$f_{w,b}(x) = \sigma(z)$$



• Good fit to the data

best one ✓

$$\begin{aligned} z = & w_1 x_1 + w_2 x_2 \\ & + w_3 x_1^2 + w_4 x_2^2 \\ & + \dots \end{aligned}$$



• the model fits the data **extremely**

oversfit (high variance)

$$\begin{aligned} z = & w_1 x_1 + w_2 x_2 \\ & + w_3 x_1^2 + w_4 x_2^2 \\ & + \dots \end{aligned}$$

- underfitting (high bias)

- too simple model, doesn't fit the training data
- Result $\rightarrow$ Poorly on both $\begin{cases} \text{training data} \\ \text{test data} \end{cases}$

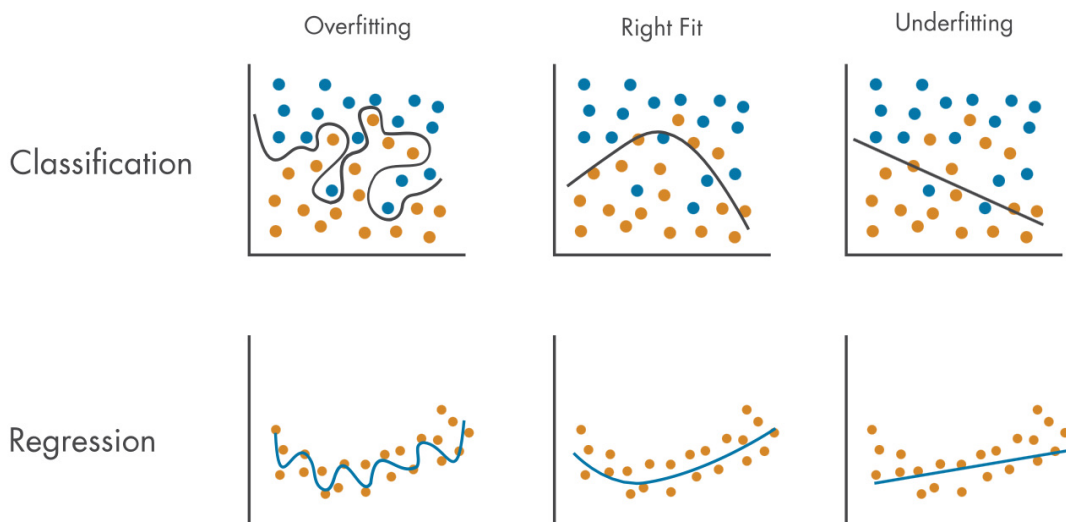
- overfitting (high variance)

- model learns too well from training data
- Result $\rightarrow$ Performs great on training data but bad on unseen / test data

- Generalization Good ✓

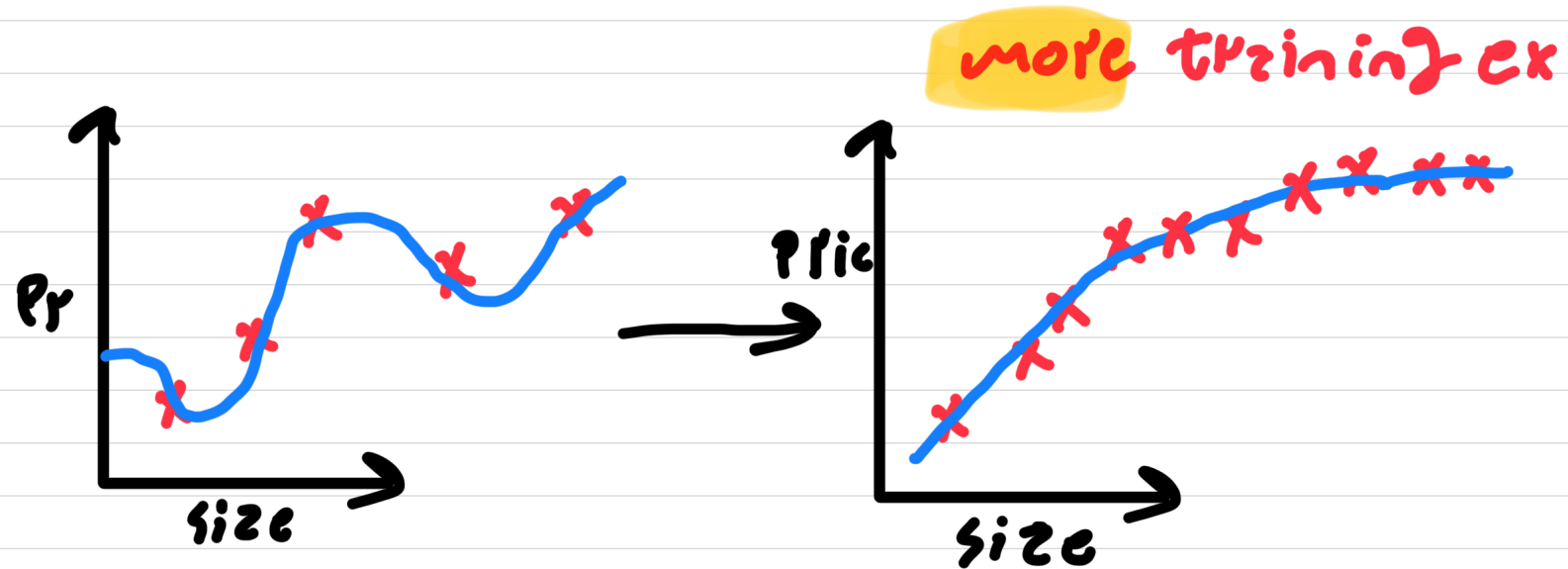
- model that performs well on new, unseen data

- Result $\rightarrow$ Balance between $\begin{cases} \text{underfitting} \\ \text{overfitting} \end{cases}$



Addressing overfitting (Handling)

1- collect more data



- Adding more training data is a solution for overfitting
- It forces the model to learn general patterns instead of memorizing noise

2-select features to include/exclude

- critical for model performance it affects overfitting, underfitting
- selecting too many features → overfitting Risk
- Too few or wrong features → underfitting Risk

size	bedrooms	floors	age	avg income	...	distance to coffee shop	price
x_1	x_2	x_3	x_4	x_5		x_{100}	y

1- All features

+

Insufficient data

↓

overfitting

2-feature selection

- size

- bedrooms

- age

If those relevant features
then we may find that the
model is no longer overfit

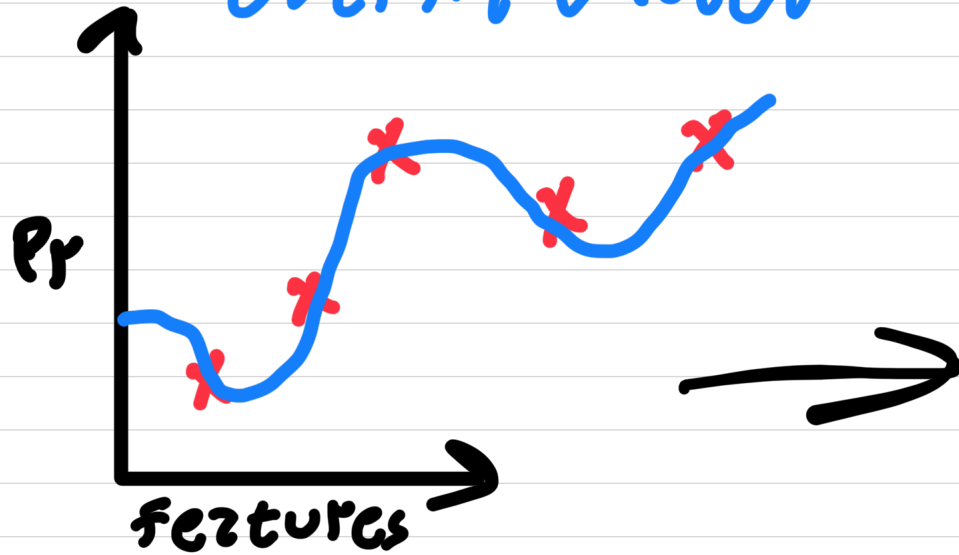
good

cons:

useful features could be lost

3- Regularization

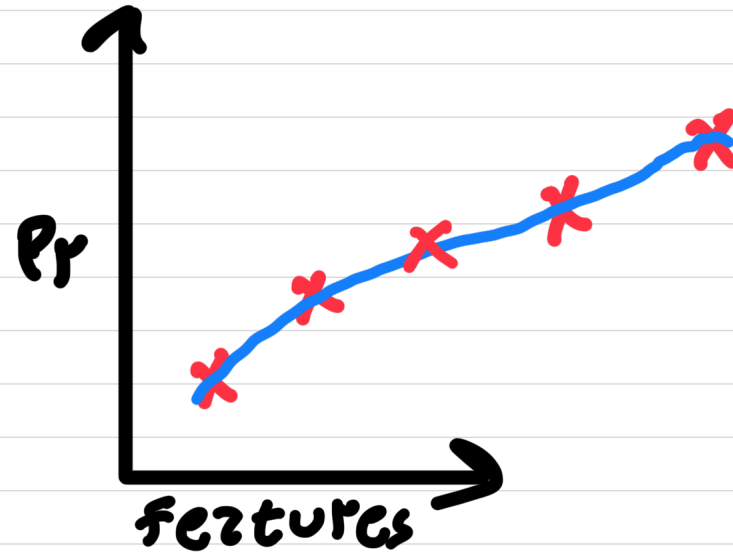
Overfit model



$$f(x) = 28x - 385x^2 + 39x^3 - 17x^4 + 100x^5$$

- large values of w_j eliminate the feature

Regularization



$$f(x) = 13x - 0.23x^2 + 0.00001x^3 - 0.0001x^4 + 10x^5$$

small w_j values

summary

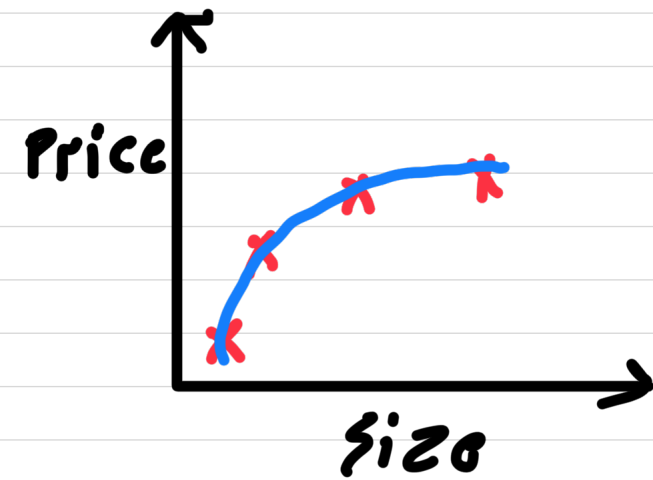
Options:

1 - collect more data Not always available

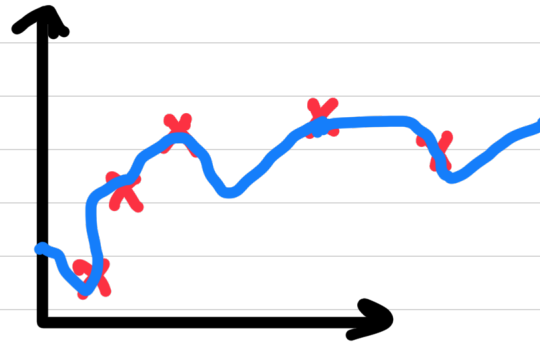
2 - select features in course 2

3 - Reduce size of parameters
"Regularization" } Next note

Regularization



$$w_1 x + w_2 x^2 + b$$



$$w_1 x + w_2 x^2 + \cancel{w_3 x^3} + \cancel{w_4 x^4} + b$$

Red squiggly lines with a zero next to them are drawn under the crossed-out terms, indicating they are approximately zero.

High order Polynomial
leads to **overfitting**

• make w_3, w_4 small ≈ 0

$$\min_{w, b} \frac{1}{2n} \sum_{i=1}^n (\hat{y} - y_i)^2 + 1000 \overbrace{w_3^2}^{0,001} + 1000 \overbrace{w_4^2}^{0,002}$$


If we want to minimize the cost function the only way is if w_3, w_4 are small $0,001, 0,002$
 $w_3 \quad w_4$

n-features

size	bedrooms	floors	age	avg income	...	distance to coffee shop	price
x_1	x_2	x_3	x_4	x_5		x_{100}	y

$w_1, w_2, w_3, \dots, w_{100}, b$ • we have 100 parameters

• we will **penalize** **all of them** by adding this term

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m (\hat{y}^i - y_i)^2 + \frac{\lambda}{2m} \sum_{j=1}^n w_j^2$$

samples $\rightarrow$ i
 features $\rightarrow$ j
 • lambda $\lambda > 0$

Summary Regularization

$$J(w, b) = \left[\underbrace{\frac{1}{2m} \sum_{i=1}^m (\hat{y} - y_i)^2}_{\text{mean squared error}} + \underbrace{\frac{\lambda}{2m} \sum_{j=1}^n w_j^2}_{\text{regularization term}} \right]$$

mean squared error

- minimize the MSE

regularization term

- minimize this term to keep w_j small which is tend to reduce overfitting
- balance both goals



$$f_{\vec{w}, b}(x) = \underbrace{w_1}_{\approx 0} x + \underbrace{w_2}_{\approx 0} x^2 + \underbrace{w_3}_{\approx 0} x^3 + \underbrace{w_4}_{\approx 0} x^4 + b$$

$$f(x) = b$$

- we want value of k that balance first and second terms

Regularized Linear Regression

$$J(\vec{w}, b) = \frac{1}{2m} \sum_{i=1}^m (\hat{y} - y_i)^2 + \frac{\lambda}{2m} \sum_{j=1}^n w_j^2$$

Annotations: "samples" with an arrow pointing to the summation index i ; "features" with an arrow pointing to the summation index j .

Gradient Descent $j = 1, \dots, n$

$$w_j = w_j - \alpha \frac{\partial}{\partial w_j} J(\vec{w}, b) \rightarrow \frac{1}{m} \sum_{i=1}^m (\hat{y} - y_i) x_i$$
$$b = b - \alpha \frac{\partial}{\partial b} J(\vec{w}, b) \rightarrow \frac{1}{m} \sum_{i=1}^m (\hat{y} - y_i)$$

Annotations: "Predicted" with an arrow pointing to $\hat{y}$; "target" with an arrow pointing to y_i .

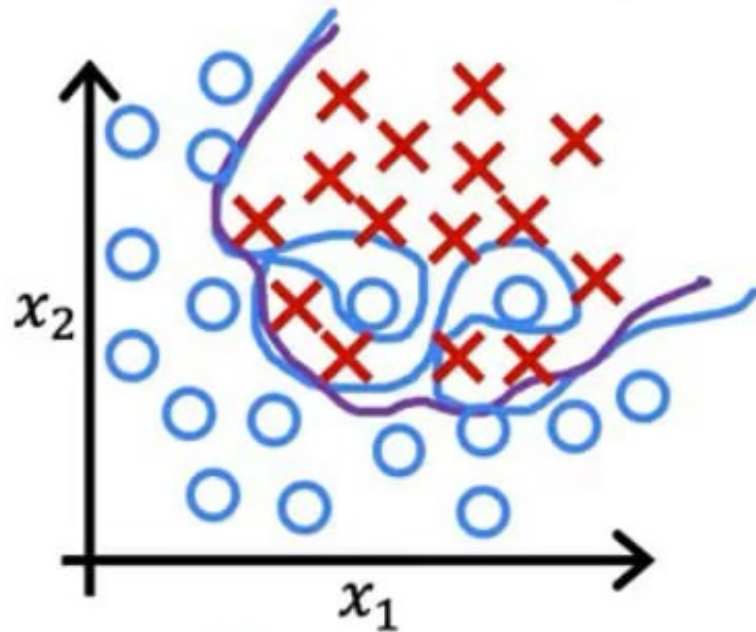
Regularized GD

$$w_j = w_j - \alpha \left[\frac{1}{n} \sum_{i=1}^n (\hat{y} - y_i) x_i + \frac{\lambda}{n} w_j \right] \quad \bullet w_j : \text{regularized}$$

$$b = b - \alpha \left[\frac{1}{n} \sum_{i=1}^n (\hat{y} - y_i) \right] \quad \bullet b : \text{don't have to regularized}$$

Regularized logistic Regression

Regularized logistic regression



$$z = w_1 x_1 + w_2 x_2 + w_3 x_1^2 x_2 + w_4 x_1^2 x_2^2 + w_5 x_1^2 x_2^3 + \dots + b$$

$$f_{\vec{w}, b}(\vec{x}) = \frac{1}{1 + e^{-z}}$$

Cost function

$$J(\vec{w}, b) = -\frac{1}{n} \sum_{i=1}^n \left[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \right] + \frac{\lambda}{2n} \sum_{j=1}^n w_j^2$$

Regularized GD

$$w_j = w_j - \alpha \left[\frac{1}{m} \sum_{i=1}^m (\hat{y} - y) x_i \right] + \frac{\lambda}{m} w_j$$

$$b = b - \alpha \left[\frac{1}{m} \sum_{i=1}^m (\hat{y} - y) \right]$$

types of Regularization

1-L2 (Ridge) $J_{L2}(w, b) = \overbrace{\frac{1}{2n} \sum_{i=1}^n (\hat{y}_i - y_i)^2}^{MSE} + \overbrace{\frac{\lambda}{2n} \sum_{j=1}^n w_j^2}^{L2}$

- Penalty = sum of squares weights
- keep all weights but shrinks them $\propto 0$

2-L1 (Lasso) : $J_{L1}(w, b) = \overbrace{\frac{1}{2n} \sum_{i=1}^n (\hat{y}_i - y_i)^2}^{MSE} + \overbrace{\frac{\lambda}{2n} \sum_{j=1}^n |w_j|}^{L1}$

- Penalty = sum of Abs values of weights
- Can Put some weights to zero

feature selection

Summary

without Regularization

the model might overfit and assign large weights even to useless features

With Regularization

- Penalize the large weights $\rightarrow$ shrinks unimportant ones to ≈ 0
- force the model to rely on useful features
- leads to simpler model